

Novel P-polarized Light Reflection Screen for AR-HUD with Wide Field of View

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Abstract

We have developed a P-polarized reflection screen for automotive windshields that suppresses ghost images in a multi-focal, wide field of view (FOV) augmented reality head-up display (AR-HUD). By incorporating a novel polarization conversion layer that compensates for polarization changes induced by phase differences across a wide FOV, the proposed technology achieves effective ghost suppression. This advancement facilitates the development of clearer AR displays in HUD, enhancing both driver safety and user convenience.

Author Keywords

Liquid crystals; Cholesteric liquid crystals; Head-up displays; Wide FOV; Reflection screen.

1. Objective and Background

Recent developments in augmented reality (AR) display technologies, including AR glasses and smart helmets, have accelerated the demand for intuitive information presentations within diverse environments. In the automotive field, AR head-up displays (AR-HUDs) that superimpose virtual content onto the windshield are rapidly advancing. Key technical requirements include expansion of the field of view (FOV), improved depth rendering through multi-focal projection, and enhanced visual clarity under varying driver's viewpoints (Figure 1).

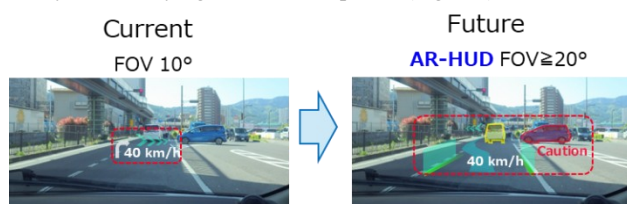


Figure 1. Conceptual image of the future AR-HUD

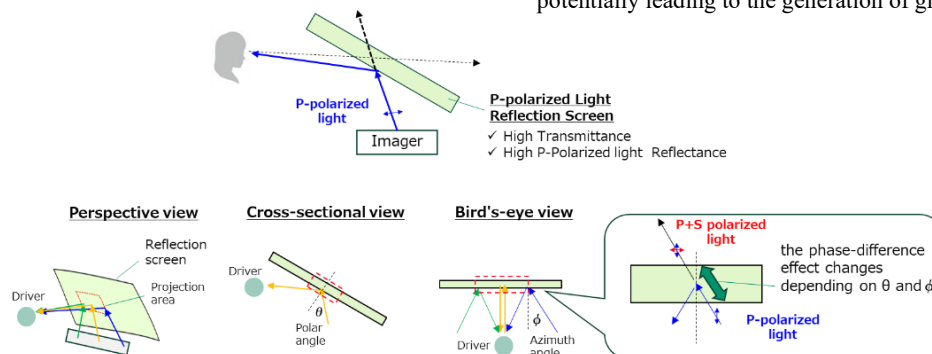


Figure 2. Requirements and issues for P-Polarized Reflection Screens

Wedge-shaped PVB is currently used in automotive HUD screens. Ghost images are eliminated by adjusting the wedge angle to superimpose the virtual images created by reflections from the inner and outer surfaces of the windshield. However, if the focal length of the virtual image or the driver's viewpoint shifts, the superimposition of the virtual images becomes misaligned, leading to the occurrence of ghost images [1]. Additionally, since S-polarized light reflection at the glass surface is utilized, HUD images become invisible when the driver wears polarized sunglasses [2].

To address these issues, solutions using P-polarized light have been proposed to suppress ghost images and ensure visibility when wearing polarized sunglasses [1,2]. Setting the angle of incidence θ of the light from the imager to the windshield to 56–65°, and using P-polarized output, the surface reflectance of glass ($n=1.5$) can be suppressed to 0–1.3% [3], thus reducing ghost images.

A P-polarized reflection screen is designed to achieve high reflectivity for P-polarized light while still maintaining the level of transmittance required by automotive windshield regulations. A common method of P-polarized reflection is to deposit thin films of inorganic and metallic materials onto the windshield [4]. However, metallic layers inherently absorb visible light, making it challenging to simultaneously achieve both high transmittance and high reflectance.

Linear polarization reflection technologies that do not absorb visible light include polymer-based multilayer films composed of alternating layers with different refractive index anisotropies [5], and cholesteric liquid crystals combined with quarter wave plates (QWP) [6]. Although these approaches do not absorb visible light, they exhibit a different limitation: The phase difference inherent in the reflective screen itself alters the polarization state of light transmitted through it. Particularly for wide-FOV HUDs, this phase-difference effect can also vary depending on the azimuth angle (ϕ) of the incident light. Consequently, the transmitted light may contain undesirable S-polarized components (Figure 2), potentially leading to the generation of ghost images.

This S-polarized light increases reflection at the outer surface of the windshield, resulting in ghost images. Therefore, suppressing ghost images requires ensuring that the light transmitted through the reflective screen remains P-polarized.

Given this background, we developed a P-polarized light reflection screen that can suppress ghost images over a wide FOV. This performance was achieved through a newly designed polarization-conversion layer (PCL) that utilizes liquid-crystal alignment technology to control light transmitted through the reflective layer into P-polarization not only in the driver's front direction but also across a wide FOV. This paper details our reflection screen.

2. Results

2-1. Design of the Polarization Conversion Layer

To validate the design of the polarization conversion layer (PCL), a linear polarization reflection screen with a QWP/CLC configuration was used as a reference structure. This configuration combines a quarter-wave plate (QWP) that converts P-polarized light to circularly polarized light and a cholesteric liquid crystal layer (CLC) that reflects circularly polarized light. PCL was applied to the side opposite QWP of CLC, forming the new configuration (Figure 3).

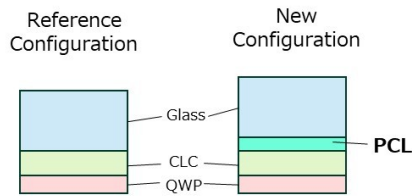


Figure 3 Sample configuration for PCL verification

The polarization state of light transmitted through the reference configuration screen is shown in Figure 4. Light transmitted through the screen becomes a mixed polarization state containing S-polarized light, which causes ghost images. To suppress ghost images, it is necessary to convert the polarization state of this transmitted light to P-polarized light.

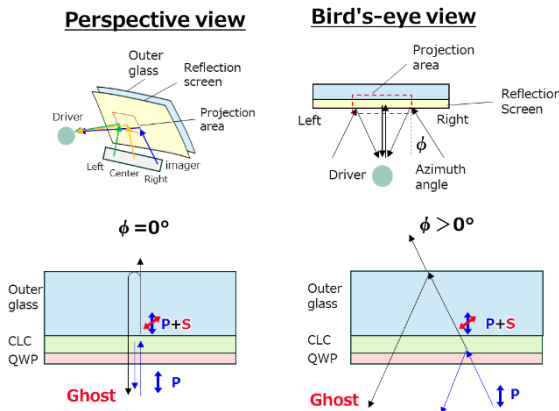


Figure 4. Image of the polarization state of transmitted light in the reference configuration.

Therefore, we devised a concept in which a PCL is incorporated to convert the transmitted light into P-polarized light across a wide FOV, thereby suppressing ghost images (Figure 5).

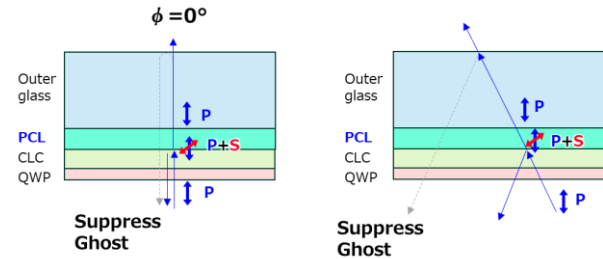


Figure 5. Concept for ghost suppression using PCL

We designed a new PCL using liquid crystal alignment technology to control the liquid crystal's $\Delta n \cdot d$ and alignment state. Figure 6 shows the simulation results for the azimuthal dependence of the P-polarized light ratio in the reference configuration (QWP/CLC) and the new configuration (QWP/CLC/new PCL). Light incident on the screen was set with a polar angle $\theta = 65^\circ$ and an azimuth angle $\phi = 0 \pm 10^\circ$, assuming a wide FOV HUD. The CLC was designed such that the average P-polarized reflectance at 450 nm to 650 nm with $\theta = 65^\circ$ is 26%.

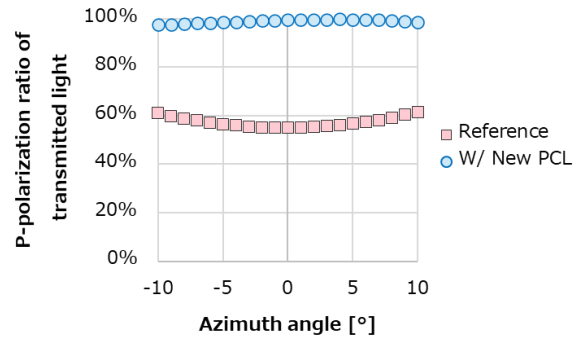


Figure 6. Simulation results of azimuth dependence of the P-polarization ratio in transmitted light.

While the reference configuration showed a decrease in the P-polarization ratio with an increasing azimuth angle, the new configuration exhibited reduced azimuth dependence. This achieved a P-polarization ratio of over 97% in calculations.

2-2. Verification of the Polarization Conversion Layer

P-polarized reflection screens were fabricated for both the reference configuration and the new configuration. The HUD image was observed by varying the azimuth angle (Figure 7). To estimate the ghost image caused by light reflecting off the outer surface of the glass after light passing through the screen, the reflection screen was laminated onto the viewing side of the glass. While the ghost images were visible with the reference configuration, no ghost images were visible with the new configuration, even as the azimuth angle changed.

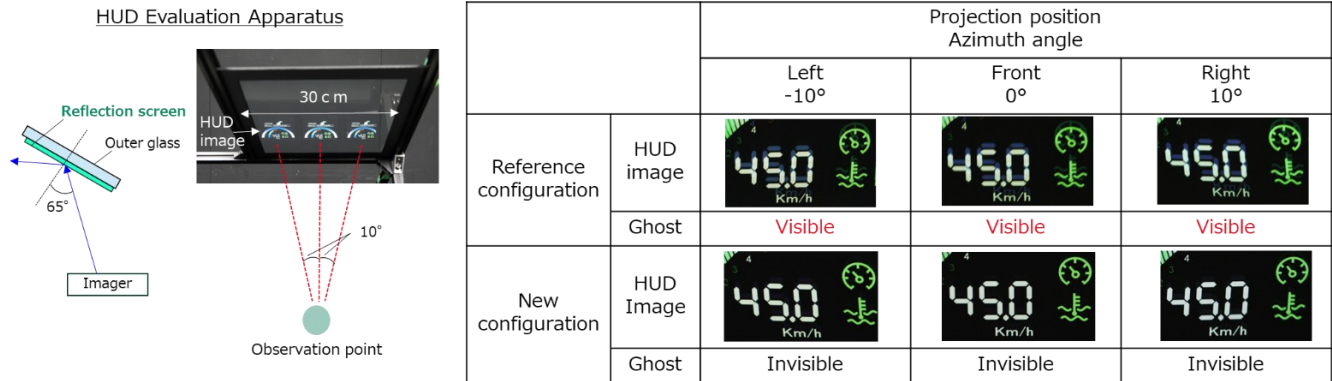


Figure 7. Verification of the ghost suppression effect with new PCL.

Furthermore, Table 1 shows the performance of a sample in which a P-polarized reflection screen was laminated inside the laminated glass. It was confirmed that the new configuration showed no difference in performance compared to the reference configuration in terms of transmittance and appearance color. Furthermore, durability evaluation showed minimal changes in P-polarized light reflectance and haze, making it suitable for automotive applications. Additionally, the P-polarized reflection screen in the new configuration can be manufactured using roll-to-roll processing and is applicable to windshield sizes (Figure 8).

Table 1. Performance of the reflective screen with new PCL.

Evaluation configuration

Evaluation Item		Angle of incidence	Reference configuration	New configuration
P-polarized light Reflectance of Screen		65°	28%	28%
Transmittance		0°	84%	84%
Appearance color (D65)		0~65°	a* -1~3 b* 4~6	a* -1~3 b* 3~6
Durability	90°C2000h	0°, 65°	ΔHaze<0.1% ΔR=0.3% ΔE=0.4	ΔHaze<0.1% ΔR=0.3% ΔE=0.4
	Xe(180W/m2) 1600h	0°, 65°	ΔHaze<0.1% ΔR=0.3% ΔE=0.5	ΔHaze<0.1% ΔR=0.3% ΔE=0.5



Figure 8. Photograph of reflective screen bulk roll and sampling image for application to the windscreen.

3. Conclusion

We designed a novel PCL that controls the polarization state of light passing through the reflection layer into P-polarized light. This demonstrates its ability to suppress the ghost images over a wide FOV. As AR-HUD technology evolves to enable virtual images to be superimposed over a wide area based on background position, we believe that applying this technology will enable clearer AR displays, expand the expressive freedom of AR-HUD, and contribute to further improvements in driver safety and convenience.

4. References

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